Luan Alfredo Lugarini

Computer Engineering Student

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Luan1357
Lattes CV

Fourth-semester Computer Engineering student at PUCPR seeking an engineering internship. Academic projects in systems programming in C (dynamic memory allocation, binary file parsing), Python and robotics (LEGO SPIKE and Arduino), including a robotic arm I assembled and programmed on my own. Advanced English (C1).

Education

2025 – 2029 **B.S. in Computer Engineering**, *Pontifical Catholic University of Paraná (PUCPR)*, Curitiba, Brazil, 4th semester, expected graduation 2029

2021 – 2023 **High School**, *Colégio Imaculada Conceição*

Academic Projects

2025 **MyArm Robotic Arm: Tower of Hanoi**, *Principles of Robotics, PUCPR*, C++ / Arduino
- Assembled on my own a 4-degree-of-freedom robotic arm (base, two joints and gripper) driven by servo motors and an Arduino.
- Programmed the motion sequence that physically solves the Tower of Hanoi puzzle, with servo speed control for smooth movement.
- Code and video: github.com/Luan1357/myarm-torre-de-hanoi

2025 **Autonomous Maze-Solving Robot**, *Principles of Robotics, PUCPR*, Python / LEGO SPIKE
- Robot that navigates a physical maze unassisted, using a distance sensor (walls), a color sensor (floor markers) and a gyroscope (90° and 180° turns with angle correction).
- Logic built as a state machine (outbound, return, finish) using asynchronous programming.
- github.com/Luan1357/robo-labirinto-lego

2026 **Conway's Game of Life**, *Imperative Programming, PUCPR*, C
- Cellular automaton simulator with a dynamically allocated grid (`malloc`/`calloc`), initial state loaded from file, and dead borders.
- Double buffering with pointer swapping between generations, avoiding data copies and incorrect in-place updates.
- github.com/Luan1357/jogo-da-vida-conway

2026 **ppm2ascii: Image to ASCII Art Converter**, *Imperative Programming, PUCPR*, C
- Full parsing and validation of binary PPM (P6) images, supporting 8 and 16 bits per channel, with overflow checks before allocation.
- Per-pixel luminance calculation mapped onto a 57-character ramp.
- github.com/Luan1357/ppm2ascii

2026 **15 Puzzle**, *Imperative Programming, PUCPR*, Python
- 4x4 sliding puzzle in the terminal, with a shuffle that always produces a solvable board and an auto-solver that replays the move history as a stack.
- github.com/Luan1357/jogo-dos-15

Technical Skills

Languages	C, Python, C++ (Arduino)
Hardware	Arduino, servo motors, LEGO SPIKE (distance, color and gyro sensors)
Tools	Git, GitHub, VS Code

Languages

Portuguese	Native
English	Advanced (C1), certified by Wise Up